Computer Networks Spring 2026
Assignment#3 (6A & 6B)

Due Date: 23:59 (11:59 pm) on Tuesday, 17th March, 2026

Submission Mode & Time: Clear ***scanned*** copy of ***handwritten*** solution in ***PDF*** format to be submitted to the ***Google Classroom (GCR)*** within the due date.

- Assignment document ***emailed*** to the ***Course Instructor*** or assignments submitted after the due time will ***not*** be accepted.
- Once ***on-campus lectures*** resume, ***original handwritten assignment solution document*** will be collected by the TA & cross matched with the PDF document submitted earlier within the deadline via email.
- In case online classes resume before the due date, then handwritten assignment solution document will be submitted during the lecture on-campus on due date.

Please note the following:

1. No exceptions to the above date and time will be allowed. Inability to submit the assignment by the required time will result in zero marks.
2. To ensure self-completion of assignments and discourage plagiarism, the instructor or the relevant TA may randomly contact you and ask for an explanation of your answers. Where plagiarism and/or cheating is evident, you will be referred to the departmental disciplinary committee. In extreme cases of plagiarism an F may be awarded immediately with further referral to the university disciplinary committee.
3. All solutions must be **hand-written** and submitted in **PDF Format.**

PART-1

Use the following text for completion of this part of the assignment:
Computer Networking - A Top-Down Approach 8th Edition by Kurose & Ross.

Solve the following problems from the back of **Chapter 3**. Every Question has equal marks i.e.

Review Questions: (3+3 = 6 marks) [CLO 1, 2]

R6, R15

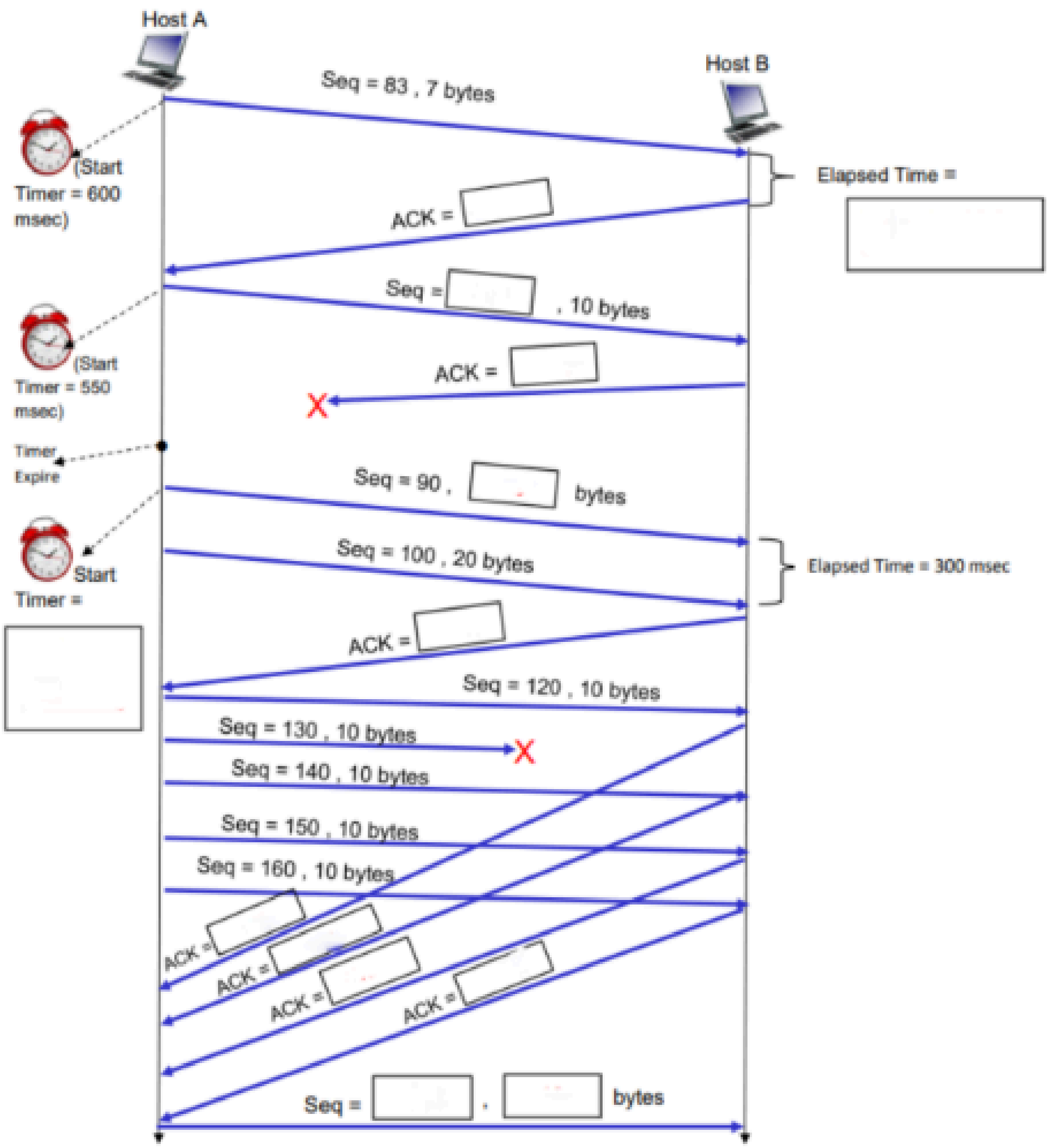
Problems: (3+3+3 = 9 marks) [CLO 2]

P2, P3, P31

PART - 2

Question: 01 [11 + 2 + 2 = 15 Marks] [CLO 1]

Refer to the TCP segments exchange between Host A to Host B. Pleasse fill in the missing values in the space provided in boxes:



Question: 02 [2 Marks] [CLO 1]

TCP generally uses a timeout mechanism to detect packet loss. However, waiting for a timeout can be inefficient. Explain the mechanism of Fast Retransmit. What specific event triggers it, and what does the sender assume has happened?

Question: 03 [2 Marks] [CLO 1]

Many Link Layer protocols (like Ethernet) already perform error checking (CRC). Why, then, does UDP (a Transport Layer protocol) implement its own checksum? Is it redundant? Explain based on the End-to-End Principle.

Question: 04 [1+2 = 3 Marks] [CLO 2]

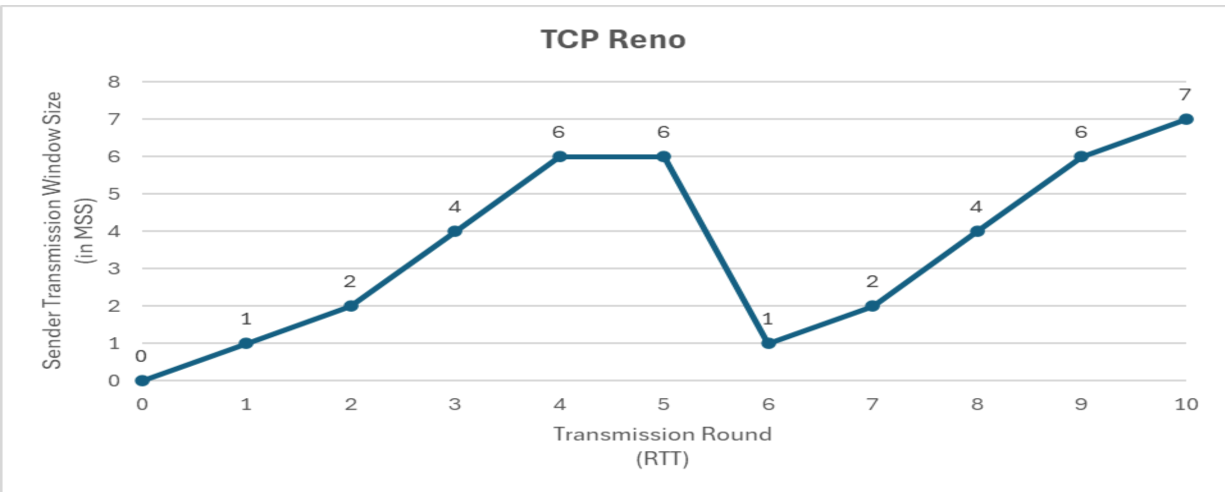
A sender is transmitting data to a receiver using sliding window protocols. Window size (N) is 5. The total number of packets to be sent are 15 (numbered as 1 to 15). Assume that if the packet is not lost, TCP receiver immediately sends its acknowledgement.

i. If selective repeat is used and packet numbered 5 and 9 are lost. How many packets will the sender retransmit in total and which ones?
ii. If Go-back-N is used and packet 6 is lost. How many packets will the sender retransmit and which ones?

Question: 05 [0.5 x 30 +2 + 0.5 x 32 = 33 Marks] [CLO 2] Consider a TCP Reno connection as illustrated by the diagram below. Please also note the points below:

- 1 MSS = 100 bytes
- At RTT = 0, TCP connection has been established but no segment has been sent by the sender yet. Initial (default) values of different parameters at RTT = 0 are given in the TCP sender table below.
- The sender starts its transmission i.e. sends the first segment with sequence number = 0, at RTT = 1
- 14th segment sent by the sender is lost
- Out of sequence received segments are not buffered but discarded
- SS is short for Slow Start, CA for Congestion Avoidance, FR for Fast Recovery & N/A for Not Applicable.

(a) For the TCP Sender table given below for this connection, some entries in the table are pre-filled. Please fill in all the remaining missing entries for this table.
(b) What is the sequence number of the segment which was lost?________________________
(c) Also, please provide a complete table with all missing rows.



TXN Round (RTT)	Receive Window (rwnd in MSS)	Congestion Window (cwnd in MSS)	Slow Start Threshold (ssthresh in MSS)	Received ACK Number	Sequence Number(s) of Transmitted Segment(s)	Timer (Timeout in ms)	No Loss Or Timeout Or 3dupAck	TCP Sender Congestion Mode (SS or CA or FR)
0	6	0	12	N/A	N/A	600	No Loss	N/A
1	6			N/A		600	No Loss	SS
2	6					600		
4	6	8				600	No Loss	
5	6		12	1300		600		
6	20							
10	20					1200		